

INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR

1. Name of the Academic Unit: Geology and Geophysics

2. Subject Name: Sedimentology Lab L-T-P: 0-0-3

Credits:2

3. Pre-requisites: None

4. Syllabus and reference books:

Syllabus:

Size analysis of sediments and determination of grain size parameters; Analysis of shape and roundness; Paleocurrent analysis; Methods of sedimentary rock identification and quantitative and qualitative analysis using hand specimens and thin sections.

Reference Books:

- 1) Principles of Sedimentology and Stratigraphy (By Sam Boggs Jr.; Pearson; Latest edition)
- 2) Sedimentology: Process and Product (By M.R. Leeder, Springer; Latest edition)
- 3) Carbonate sedimentology (By Maurice E. Tucker, & V. Paul Wright; Wiley, Latest edition)
- 4) Depositional Sedimentary Environments (by H. E. Reineck and Indra Bir Singh; Springer-Verlag Berlin and Heidelberg GmbH & Co. Latest edition)
- 5) Sedimentology and Petroleum Geology (by Knut Bjorlykke, Springer)

5. Lecture-wise break-up:

Sl. No.	Topic	No. of lectures
1.	Size analysis of sediments and determination of grain size parameters, Analysis of shape and roundness, Paleocurrent analysis	2
2.	Sedimentary rock identification using hand specimens	1
3.	Thin sections study: Quantitative grain size analysis	1
4.	Thin sections study: Identification of sedimentary rocks (Siliciclastic and Carbonate)	2
5	Thin sections study: Understanding matrix and cement	2
6	Diagenesis of sediments; Siliciclastic	2
7	Diagenesis of sediments-Carbonates	2
Total number of hours		12